Bayesian inference - calculate probability following chains of dependencies or correlations

Bayes' Theorem (conditional probability)

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

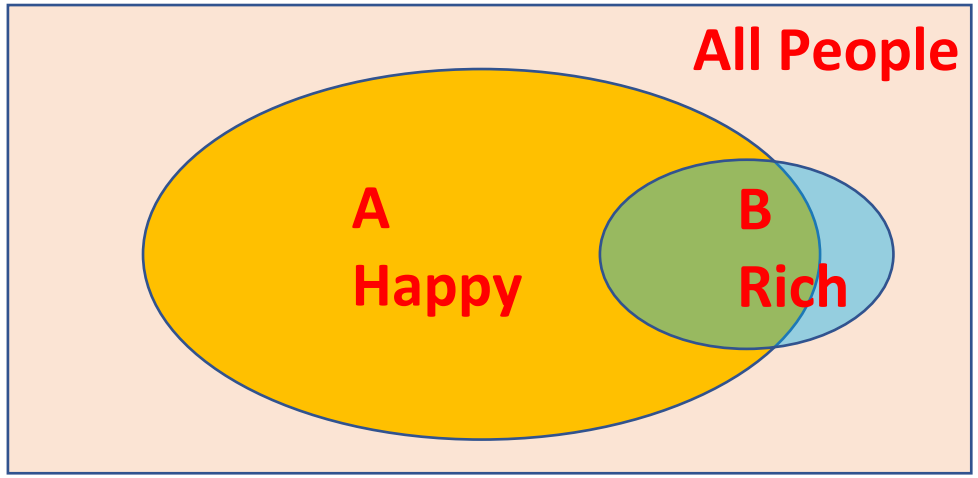
or

$$P(A|B) * P(B) = P(B|A) * P(A)$$

where A and B are events, and P(B) not equal 0.

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{evidence}}$$

Thomas Bayes
(1701–1761)



Intuitive explanation of Bayesian rule (on the left).
A = Happy, B = Rich
P(A) = 40% = probability of being happy (Yellow)
P(B) = 5% = probability of being rich (Blue)
P(A & B) = 4% = probability of rich & happy (Green)
P(B|A) = 10% = probability of being rich given that Happy
P(A|B) = 80% = probability of being happy given that rich

You can calculate green intersection using two methods:
P(A|B)*P(B) = 80%*5% = 4%
P(B|A)*P(A) = 10%*40% = 4%

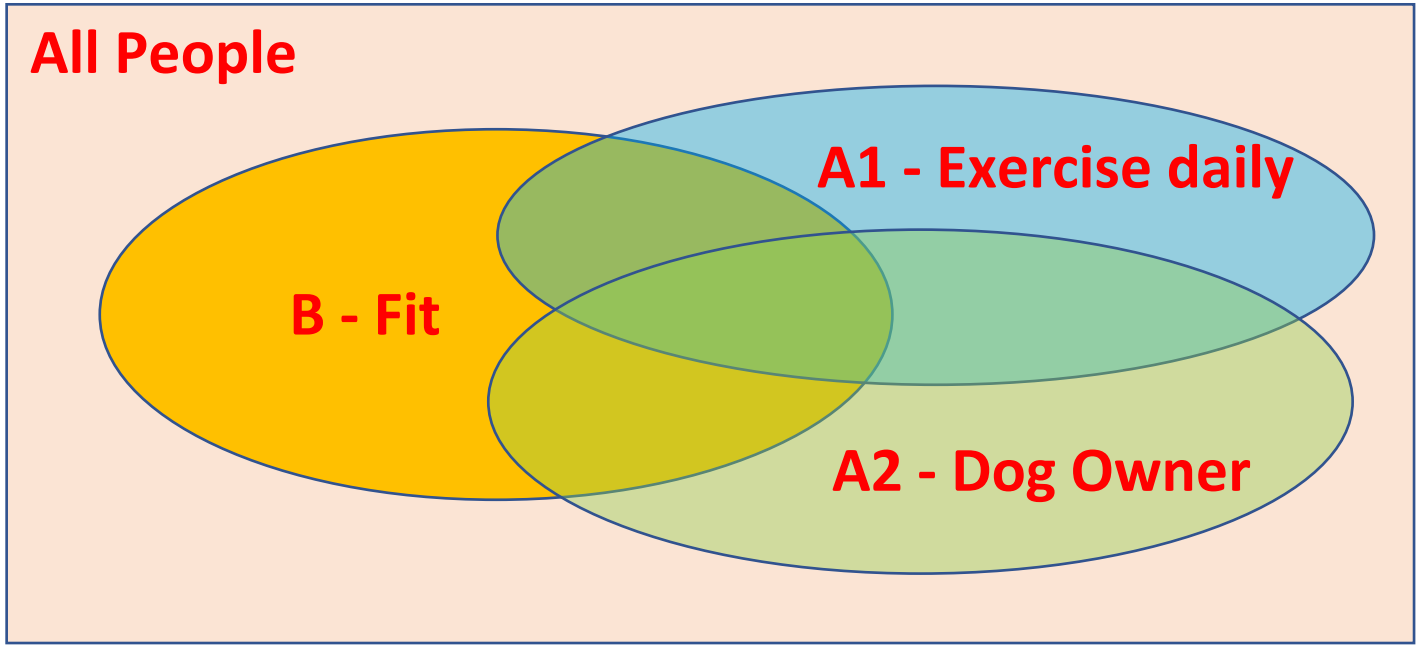
Naive Bayes Classifiers

Simple probabilistic classifiers based on applying Bayes' theorem.
The main "Naive" assumption is that although individual features may have inter-dependencies and may have different importance, we "naively" assume that each feature makes an **independent** and **equal** contribution to the outcome.

Bayes' Theorem

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

On the picture below we have two features (A1 & A2) influencing outcome (B).
We will be computing conditional probability of a person being fit P(B|(A1,A2)).
We "naively" assume that A1 & A2 contribute equally to the outcome.
But in reality daily exercising probably contributes more.
Our naive model doesn't take this into account.



Features:
- A1 – Exercise Daily
- A2 – Dog owner

Outcome:
- B – Person is Fit

Naive Bayes classifiers - continued

In the table to the right we have 4 columns A_i with features and one outcome column B - "Play Golf".
For each value of the outcome (Yes or No) we can calculate the conditional probability/frequency of seeing each feature value A_i for a given outcome B:

$$P(A_i | B)$$

For example we can calculate how often we played or not played golf when it was rainy, etc.
Then we will be calculating the opposite – probability of playing golf given a specific set of features: $P(B|A_i)$. We will use Bayes formula for this.

In other words, we will be able to predict outcome B from features A_i.

A_i B

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Here is a very good short video explaining the Naive Bayes Classification on a specific simple example:
- https://www.youtube.com/watch?v=CPqOCI0ahss
Note – there is a (common) mistake in this video! Author calculates denominators via multiplication of individual probabilities. This is wrong. You can see it by yourself how making calculation this way makes the sum of two probabilities greater than one.
Also here is a good article - https://www.geeksforgeeks.org/naive-bayes-classifiers/

Naive Bayes Classification in more details:

Suppose that an outcome "O" can have values **True** or **False** (Binary Classification) depending on events e1, e2, e3.
Suppose that influences of these events on the outcome are mutually independent.
Then we can write Bayes' Theorem formula like this:

$$P_{true} = P(True|e1,e2,e3) = \frac{P(e1,e2,e3|True) * P(True)}{P(e1,e2,e3)} = \frac{P(e1|True) * P(e2|True) * P(e3|True) * P(True)}{P(e1,e2,e3)}$$

and similarly:

$$P_{false} = P(False|e1,e2,e3) = \ldots = \frac{P(e1|False) * P(e2|False) * P(e3|False) * P(False)}{P(e1,e2,e3)}$$

Note how we write conditional probability P(e1,e2,e3|O) on top as a product of individual probabilities.
We can do that because their influences are presumed independent and equal (this is the main assumption of "Naive" approach).

Also note that the probability in the denominator is NOT split into multiplication.
We can not split it because generally we don't assume that events e1,e2,e3 are independent.
We only assume that their influences on the outcome "O" are independent (and equal in their potential influence).

But the good news is that we don't have to calculate the denominator, because it is the same for both probabilities, and for classification purposes we only need to determine which value is larger – P_{true} or P_{false}.

Also note that if you want to use this formula for only few events (for example e1 & e2), you can simply remove all other events from the formula as if you have never known that they existed. Ultimately for just one event e1 you will get classical Bayes formula:

$$P(O|e1) = \frac{P(e1|O) * P(O)}{P(e1)}$$

Frequentist interpretation of Bayes' theorem.

Probability measures a proportion of outcomes.
P(A) is the proportion of outcomes with A, P(B) – with B.
P(B|A) is the proportion of outcomes with property B
out of outcomes with property A.

Bayesian interpretation of Bayes' theorem.

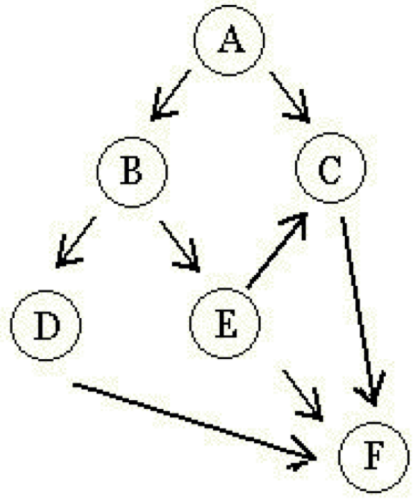
Probability measures a degree of belief.
P(A), the **prior**, is the initial degree of belief in A.
P(A|B), the **posterior**, is the degree of belief having accounted for B.
P(B|A)/P(B) – represents the support B provides for A.

Bayesian optimization

A sequential design strategy for global optimization of black-box functions that doesn't require derivatives.

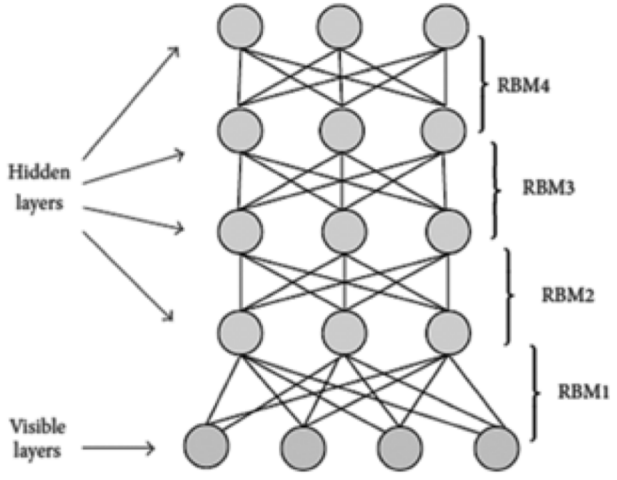
Bayesian network

(Bayes network, belief network, Bayes(ian) model or probabilistic directed acyclic graphical model) is a probabilistic graphical model (a type of statistical model) that represents a set of random variables and their conditional dependencies via a directed acyclic graph (DAG).



Example:

Deep Belief Networks - made of layers of RBMs
(Restricted Boltzmann Machines),
each next layer ("B") is trained on top of the previous layer ("A").



Bayesian vs Frequentist

Allen Downey – https://www.youtube.com/watch?v=R6d-AbkhBQ8

Bayesian:

Enumerate all possible hypotheses – and compute their probabilities.
We make a (subjective) choice of “prior” (model, distribution)
and produce posterior distribution.
Allows for complex decision analysis under uncertainty.

Frequentist (classical):

Calculate means (estimates) and confidence intervals
Problems treating missing data (survivorship bias)
Problems with hypothesis testing (p-hacking)

Example1: Polling, two candidates: Alice & Bob.
52% of polled voters intend to vote for Alice.
Confidence interval [42%,62%], p-value = 0.34
What is the probability that Alice will win ?
Classical frequentist approach can not answer this question.
Bayesian method – result is a posterior distribution of possible outcomes based on data.
Thus you can get your answer.

Example2: comparing two drugs A & B.
A is better than B with p value < 0.001
A is more expensive.
Which drug should we prescribe?
Cost-benefit.
Suppose we get predicted probability of survival with A and B.
We could then compute dollars per life saved.

Example3: Guessing game
You see the prize – and guess the price
Opponent sees a different prize – and guesses the price.
Whoever is closer (without going over) wins.
comparing two drugs A & B.

Example4: MCMC (Markov-Chain Monte-Carlo)

Bayesian vs Frequentist

Jake VanderPlas
- https://www.youtube.com/watch?v=KhAUfqhLakw

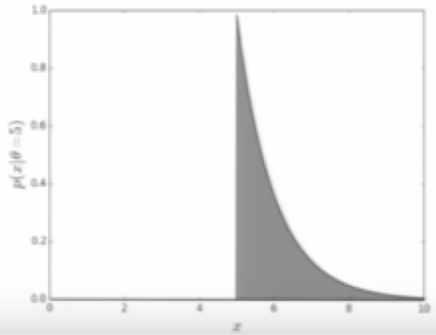
Frequentist approach gives interval 10.2 - 12.2
Bayesian approach gives interval 9.0 - 10.0

Example: estimating θ

Uncertainties: Jaynes' Truncated Exponential

Consider a model: $p(x \mid \theta) = \begin{cases} \exp(\theta - x) & , \quad x > \theta \\ 0 & , \quad x < \theta \end{cases}$

We observe D = {10, 12, 15}
What are the 95% bounds on Θ?



Jaynes 1976

Frequentist Approach

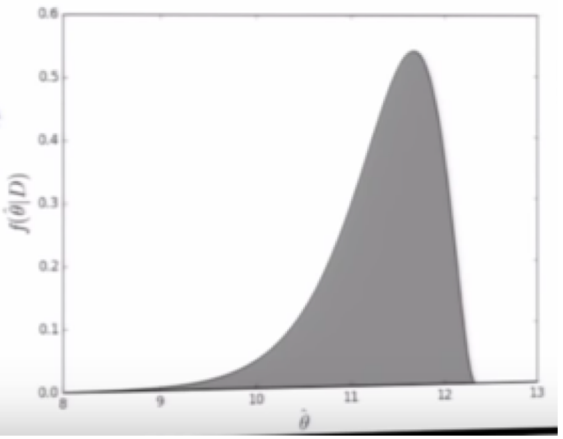
The expectation of x is:

$$E(x) = \int_0^\infty x p(x) dx = \theta + 1$$

So an unbiased estimator is:

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^{N} x_i - 1$$

Now we compute the *sampling distribution of the mean* for p(x):

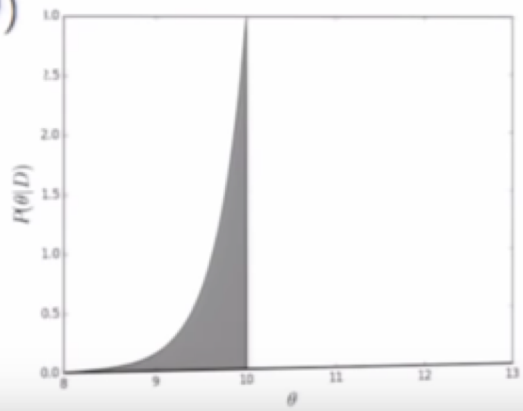


Bayesian Approach

Bayes' Theorem:

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Likelihood:

$$P(D|\theta) \propto \prod_{i=1}^{N} P(x_i|\theta)$$

With a flat prior, we get this posterior: